Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Real Estate SaaS · Backend Services & Data Pipelines · Full-Stack · Production Reliability · 9yr

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PROFESSIONAL SUMMARY

Software engineer with 9 years owning production systems for the real estate industry -- the same domain Zillow serves. At Entrata, I was the long-term sole owner of compliance-critical backend services and data pipelines for rental housing: the federal income-limit import that determines eligibility for 100K+ housing units, the async processing system for rent calculations across thousands of tenant records, and full-stack modules used by property managers across the country. I deliver end-to-end: design, build, test, deploy, monitor, and fix at 2am when things go wrong. I know real estate data -- leasing, onboarding, eligibility, recertification -- from 8 years of production ownership. I mentor engineers, lead code reviews, and build the internal tooling that makes teams faster. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Portfolio: venkatalakshmiparimala.github.io/Portfolio/about.html

TECHNICAL SKILLS

Languages & Frameworks: Python (9yr production), TypeScript, JavaScript, SQL/PL-SQL, R, Bash/shell scripting, HTML/CSS; FastAPI, Flask, Angular, React.js; REST APIs, GraphQL (familiarity), gRPC (familiarity)

Backend, Data & Systems Design: PostgreSQL (200+ queries optimized, stored procedures, multi-tenant schemas, OLTP/OLAP), MySQL, Oracle, NoSQL; ETL/ELT Pipelines, Data Modeling, Microservices Architecture, Distributed Systems, Scalable Maintainable Backend Services, System Design, RabbitMQ async processing, Event-Driven Systems, Service Reliability, Observability, Performance Engineering, Security & Data Privacy, API Design, Contract-Driven Development

Cloud, Infrastructure & Engineering Practices: AWS (S3, EC2, RDS, Glue), Docker, Kubernetes, CI/CD (GitHub Actions), Automated Testing, Unit Testing, Integration Testing, Code Reviews, Git, Agile/Scrum, Production Monitoring, Incident Response; AI/ML tooling: LangGraph, LLM APIs, RAG Pipelines, LLM Evaluation

EXPERIENCE

Entrata -- Multifamily Real Estate SaaS (PropTech)

Sep 2018 -- Aug 2024

Senior Software Engineer -- Real Estate Platform, Backend Services & Data Pipelines · Lehi, UT (Remote)

- **8 years owning the systems that power real estate workflows -- the same domain Zillow's mortgage and transaction platform serves.** Long-term sole owner of Entrata's federal income-limit import pipeline (annual ingestion of complex federal data → ~180,000 domain records powering eligibility, rent, HAP, and tax-credit calculations for 100K+ housing units). Pre-import validation, reconciliation, and rollback preparation on every release -- a wrong record means thousands of residents get the wrong eligibility decision. Zero compliance violations across the full ownership period.
- **Scalable, reliable backend services and data pipelines at production scale.** RabbitMQ-based async processing for gross-rent updates across thousands of tenant records -- designed retry logic and idempotency guarantees eliminating double-processing failures. Optimized 200+ SQL queries (30% performance gain) through execution plan analysis, indexing, partitioning, and stored procedure refactoring. Multi-tenant schemas for eligibility, financial, and document workflows. AWS S3 migration with Libsodium-encrypted ETL pipelines (100% PHI compliance, zero security incidents).
- **Full-stack delivery and operational excellence.** Shipped 20+ full-stack production modules (Python, Angular/JavaScript, PostgreSQL) across 120+ Agile sprints; Docker microservices redesign (50% scalability improvement, 30% deployment time reduction); eliminated 80% of manual operations workload through automation pipelines; maintained 99.5% uptime across compliance-critical systems.
- **Engineering culture and mentorship.** Mentored 5+ engineers; conducted 300+ peer code reviews; built internal debugging tools reducing debugging time 40%; kept bug count below 10 for 6+ consecutive months; championed testability and CI/CD standards; active mentoring across the full engineering team.

George Mason University

Aug 2024 -- May 2026

Research Associate & Graduate Teaching Assistant

- Built multi-source ML prediction pipelines (Python, R) across 190+ countries (R^2=0.977); applied Isolation Forest to automate data integrity checks across 3 research pipelines; designed Tableau/ggplot2 dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and data engineering.

Magizhs Tech Solution

May 2025 -- Aug 2025

AI/ML Engineer -- Backend Systems & API Integration

- Built backend agentic system with LangGraph orchestration, RAG retrieval, and validation guardrails; designed multi-step API integration pipeline with conditional routing and graceful failure handling; enforced regulatory constraints before export.

Entrata India -- Pune

Jan 2017 -- Sep 2018

Associate Software Engineer -- Python · PostgreSQL · Real Estate Platform

- Built property management and compliance modules; income-limit import for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints; 56 state-specific housing certification templates.

PROJECTS

Leasel: Production Backend Platform -- Scalable Services, APIs & Observability

FastAPI · LangGraph · Python · PostgreSQL · CI/CD · Production AI

Python · LangGraph · FastAPI (15-endpoint REST API) · TF-IDF RAG · SQLite · OCR · GitHub Actions | github.com/venkatalakshmiparimala/Lease_complaince_rag

Built a production backend platform from design to deployment: 15-endpoint FastAPI REST API, multi-agent orchestration pipeline (8 services with conditional routing and observability), CI regression gate blocking quality degradation, structured logging for incident triage, and HITL escalation for high-risk decisions. Demonstrates the end-to-end ownership pattern -- design, code, test, deploy, monitor -- that scales to complex production systems. 93.6% precision across 1,240 test cases.

AgentProbe: CI-Integrated Quality Framework -- Automated Testing & Regression Detection

Automated Testing · CI/CD · Observability · FastAPI · React · TypeScript

Python · FastAPI · React.js · TypeScript · GitHub Actions · ChromaDB | github.com/venkatalakshmiparimala/AgentProbe

Automated testing and observability framework: CI-integrated regression gate (GitHub Actions), 5-metric quality engine, structured tracing showing what data was retrieved and where failures occurred, multi-provider API routing. Built to enforce code quality and catch regressions before they reach production -- the same engineering practice discipline Zillow's platform requires.

AI on FHIR: Full Stack REST API Platform -- Flask & React

Flask · React · TypeScript · REST API · Full Stack · HIPAA

Python · Flask · React · Docker · spaCy · BERT · FHIR API | github.com/venkatalakshmiparimala/AI_on_FHIR

HIPAA-compliant Flask + React full-stack platform -- REST API design, RBAC, containerized deployment; reduced manual retrieval time ~60%. Full-stack delivery: Python backend, React/TypeScript frontend, Docker.

EDUCATION

M.S. Data Analytics Engineering -- George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2013--2017